WORKED SOLUTION

STEP III 2003, Question 12

Probability and Statistics • Exponential Waiting Times

STEP III 2003, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q12

The key idea

If the number of interruptions in any interval is Poisson, then "the next interruption is more than t minutes away" is the same event as "there are *no* interruptions in the next t minutes", and that turns a discrete distribution into a continuous one: the waiting time is exponential. After that, the successive gaps between interruptions are independent, so the search for a long gap is a sequence of independent trials, each a "success" (a gap of half an hour or more) or a "failure". Geometric distributions and conditional expectation do the rest.

Solution

The distribution of T

Interruptions occur on average every ten minutes, so the rate is $\frac{1}{10}$ per minute and the number of interruptions in a period of t minutes is Poisson with mean $\frac{t}{10}$.

Start the clock at the interruption that has just occurred. The time T to the next one is at least t exactly when there are no interruptions in the next t minutes, so

$$\mathrm{P}(T \geq t) = \mathrm{P}(\text{no interruptions in time } t) = \mathrm{e}^{-t/10}.$$

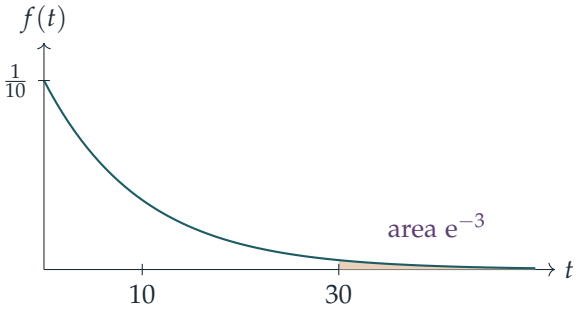
Hence, for $t \geq 0$, the probability of less than t minutes of work is

$$\mathrm{P}(T < t) = 1 - \mathrm{e}^{-t/10}.$$

This is the cumulative distribution function $F(t)$ of T (with $F(t) = 0$ for $t < 0$). Differentiating,

$$f(t) = F'(t) = \tfrac{1}{10}\,\mathrm{e}^{-t/10} \quad (t \geq 0), \qquad f(t) = 0 \quad (t < 0).$$

So T is exponential with mean 10 minutes.



The expected number of interruptions

Because the process has no memory (the Poisson counts in non-overlapping intervals are independent), each time an interruption occurs the time to the next one starts afresh with the same distribution as T, independently of everything before. So the successive gaps $T_1, T_2, \ldots$ are independent, each distributed like T.

Call a gap *long* if it is at least 30 minutes. Each gap is long with probability

$$p = \mathrm{P}(T \geq 30) = \mathrm{e}^{-30/10} = \mathrm{e}^{-3},$$

independently of the others. Let N be the number of interruptions before the first long gap (not counting the one that has just happened). Every short gap ends in an interruption and the first long gap is my uninterrupted half hour, so N is the number of short gaps before the first long one. Hence

$$\mathrm{P}(N = n) = (1-p)^n p, \qquad n = 0, 1, 2, \ldots$$

? **Ask yourself** How do we find the mean of this geometric distribution, which starts at 0 rather than 1?

Differentiate the geometric series $\sum_{n\geq 0} x^n = \frac{1}{1-x}$ to get $\sum_{n\geq 1} n x^{n-1} = \frac{1}{(1-x)^2}$ for $|x| < 1$. With $x = 1 - p$,

$$\mathrm{E}(N) = \sum_{n=1}^{\infty} n(1-p)^n p = p(1-p) \cdot \frac{1}{p^2} = \frac{1-p}{p}.$$

(Alternatively, $N + 1$ is the number of trials up to and including the first success, which has mean $\frac{1}{p}$.) With $p = \mathrm{e}^{-3}$,

$$\mathrm{E}(N) = \frac{1 - \mathrm{e}^{-3}}{\mathrm{e}^{-3}} = \mathrm{e}^3 - 1,$$

as required.

The expected length of a short gap

We want $\mathrm{E}(T \mid T < 30)$. The conditional density of T given $T < 30$ is $f(t)/\mathrm{P}(T < 30)$ on $0 \le t < 30$, so

$$\mathrm{E}(T \mid T < 30) = \frac{1}{1 - \mathrm{e}^{-3}} \int_0^{30} \frac{t}{10}\, \mathrm{e}^{-t/10}\, \mathrm{d}t.$$

Integrating by parts with $u = t$ and $\mathrm{d}v = \frac{1}{10}\mathrm{e}^{-t/10}\,\mathrm{d}t$, so $v = -\mathrm{e}^{-t/10}$,

$$\begin{aligned}\int_0^{30} \frac{t}{10}\, \mathrm{e}^{-t/10}\, \mathrm{d}t &= \Big[-t\, \mathrm{e}^{-t/10} \Big]_0^{30} + \int_0^{30} \mathrm{e}^{-t/10}\, \mathrm{d}t \\ &= -30\, \mathrm{e}^{-3} + 10(1 - \mathrm{e}^{-3}) \\ &= 10 - 40\, \mathrm{e}^{-3}.\end{aligned}$$

Therefore, multiplying top and bottom by e^3,

$$\mathrm{E}(T \mid T < 30) = \frac{10 - 40\, \mathrm{e}^{-3}}{1 - \mathrm{e}^{-3}} = \frac{10\, \mathrm{e}^3 - 40}{\mathrm{e}^3 - 1} \text{ minutes} \approx 8.43 \text{ minutes}.$$

It makes sense that this is a little less than 10: we have thrown away the long gaps.

The expected wait

The wait before the uninterrupted half hour begins is the sum of the N short gaps. Given $N = n$, these are n independent gaps each conditioned to be shorter than 30 minutes, so the expected wait given $N = n$ is $n\,\mathrm{E}(T \mid T < 30)$. Averaging over N,

$$\mathrm{E}(\text{wait}) = \mathrm{E}(N)\, \mathrm{E}(T \mid T < 30) = (\mathrm{e}^3 - 1) \cdot \frac{10\, \mathrm{e}^3 - 40}{\mathrm{e}^3 - 1} = 10\, \mathrm{e}^3 - 40 \text{ minutes},$$

which is about 161 minutes, or 2 hours 41 minutes. (If you want the time until the paper is *finished*, add the half hour itself: $10\, \mathrm{e}^3 - 10 \approx 191$ minutes.)

Checks

- $\int_0^{\infty} \frac{1}{10}\mathrm{e}^{-t/10}\, \mathrm{d}t = 1$, so f is a genuine density, and its mean is 10, matching "on average every ten minutes".
- Unconditionally, $\mathrm{E}(T) = 10$ should split as $\mathrm{P}(T < 30)\, \mathrm{E}(T \mid T < 30) + \mathrm{P}(T \ge 30)\, \mathrm{E}(T \mid T \ge 30)$. By memorylessness $\mathrm{E}(T \mid T \ge 30) = 40$, and indeed $(10 - 40\mathrm{e}^{-3}) + 40\mathrm{e}^{-3} = 10$.
- A simulation of 200 000 runs of the process gave an average of 19.1 interruptions (against $\mathrm{e}^3 - 1 \approx 19.09$) and an average wait of 161.1 minutes (against $10\mathrm{e}^3 - 40 \approx 160.86$).

Common slips

- Using $\mathrm{E}(N) = \frac{1}{p} = \mathrm{e}^3$: that counts the long gap as well, but it ends with no interruption before the half hour.

- Multiplying $\mathrm{E}(N)$ by the unconditional mean 10 to get the wait. The gaps that make up the wait are known to be short, so their mean is $\mathrm{E}(T \mid T < 30)$, not 10.

Beyond STEP

The interruptions form a *Poisson process*, and the result that the gaps are independent exponentials is one of its defining properties; the final step, $\mathrm{E}(\text{sum of } N \text{ terms}) = \mathrm{E}(N) \times \mathrm{E}(\text{term})$, is a conditioning argument closely related to *Wald's identity*. Together they lead into renewal theory and queueing models, studied in first- and second-year probability courses. Grimmett and Welsh, *Probability: An Introduction*, covers the Poisson process and conditional expectation at the right level.

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